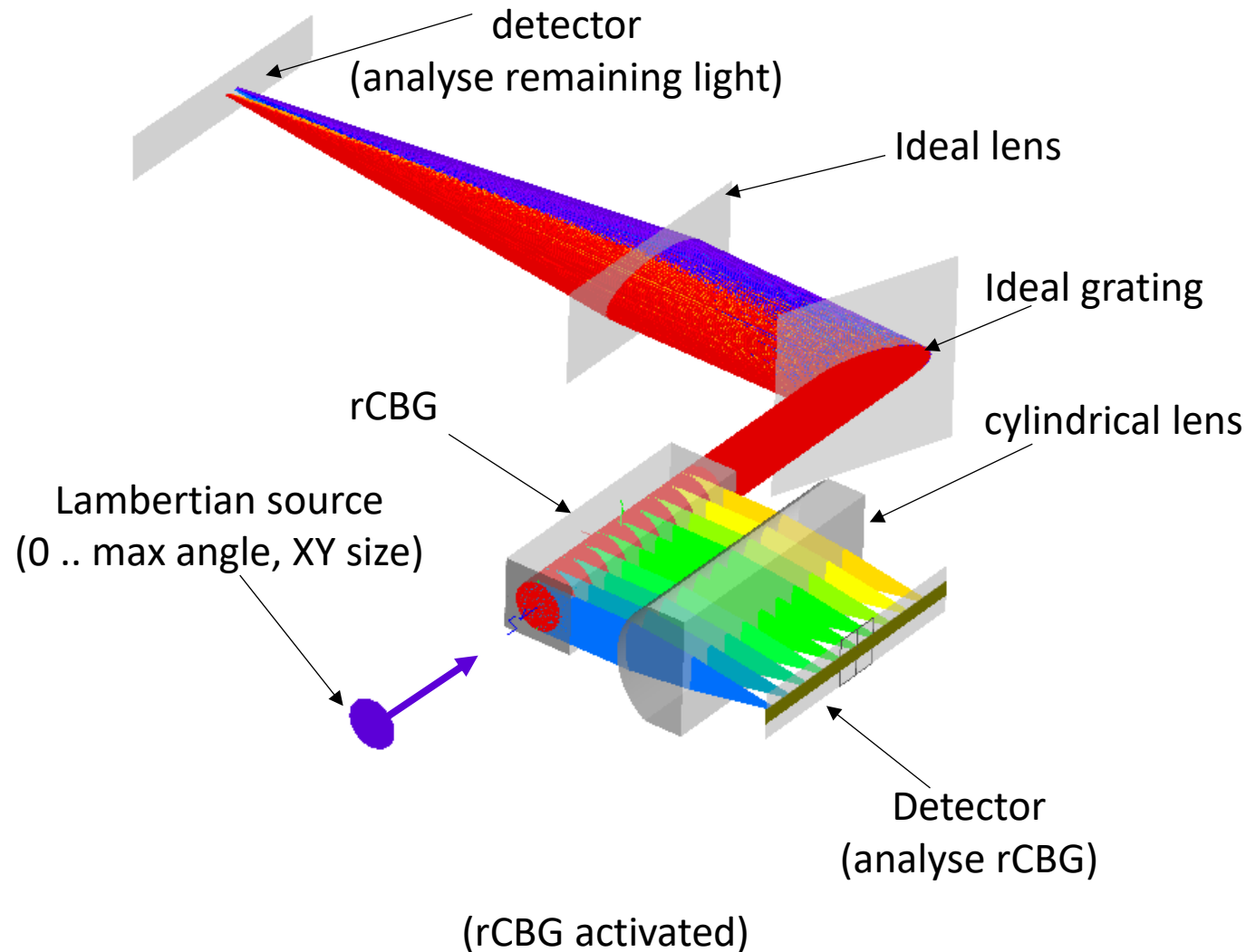




Rotated Chirped Bragg Grating implementation in FRED

Daniel Csati, 20.04.2026



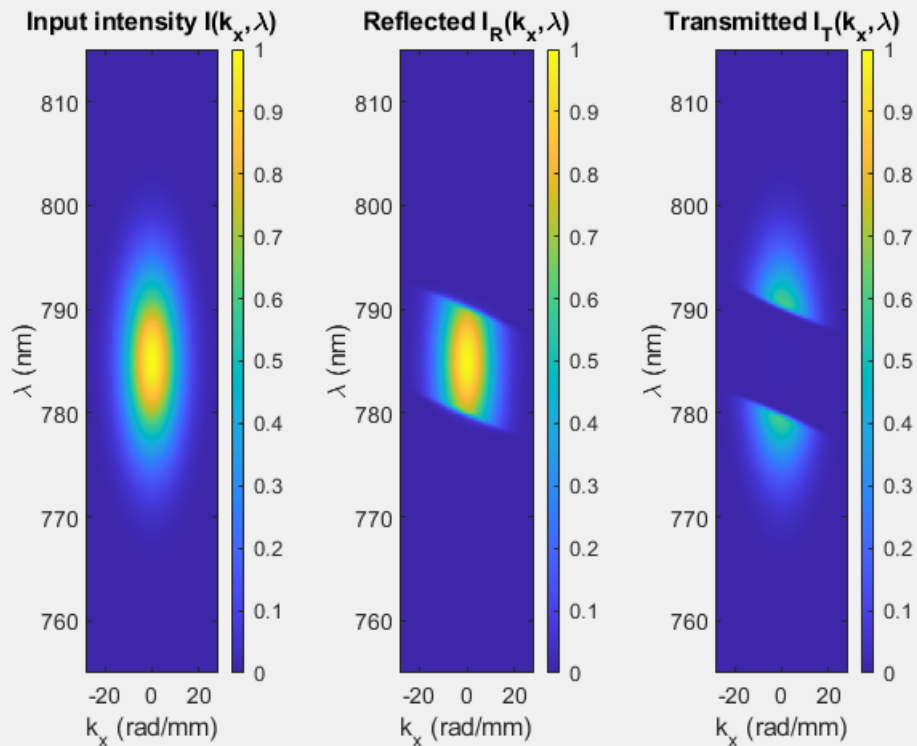
- We have created center wavelength and bandwidth plots with input ray angle and wavelength as input variables.
- We demonstrated that the simulation is working with arbitrary source and giving results as expected.

Open points:

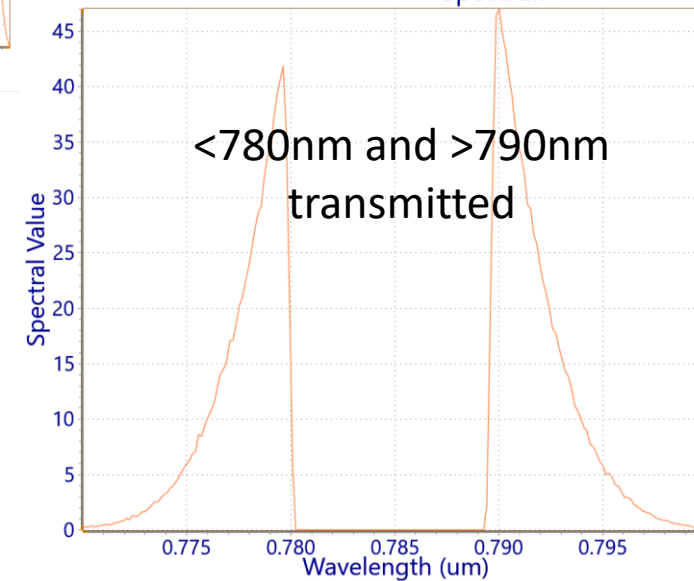
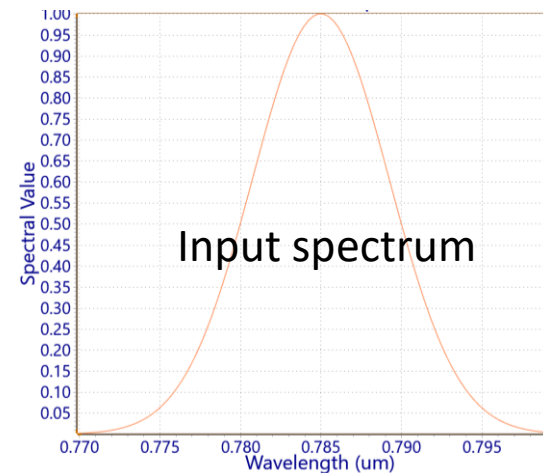
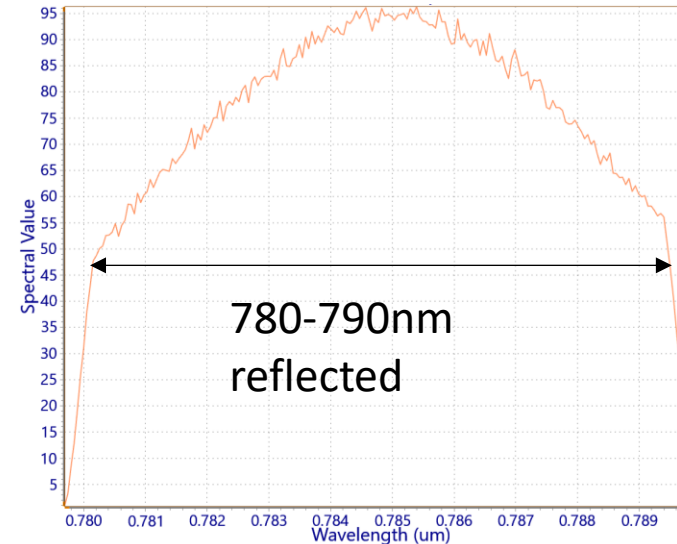
1. Comparison of MATLAB and FRED output
2. Add resolution limit imitating wave-optics-like behaviour.
3. Debugging and clean-up FRED script.
4. Work instructions / documentation.

MATLAB model

Spectral domain (k_x, λ)

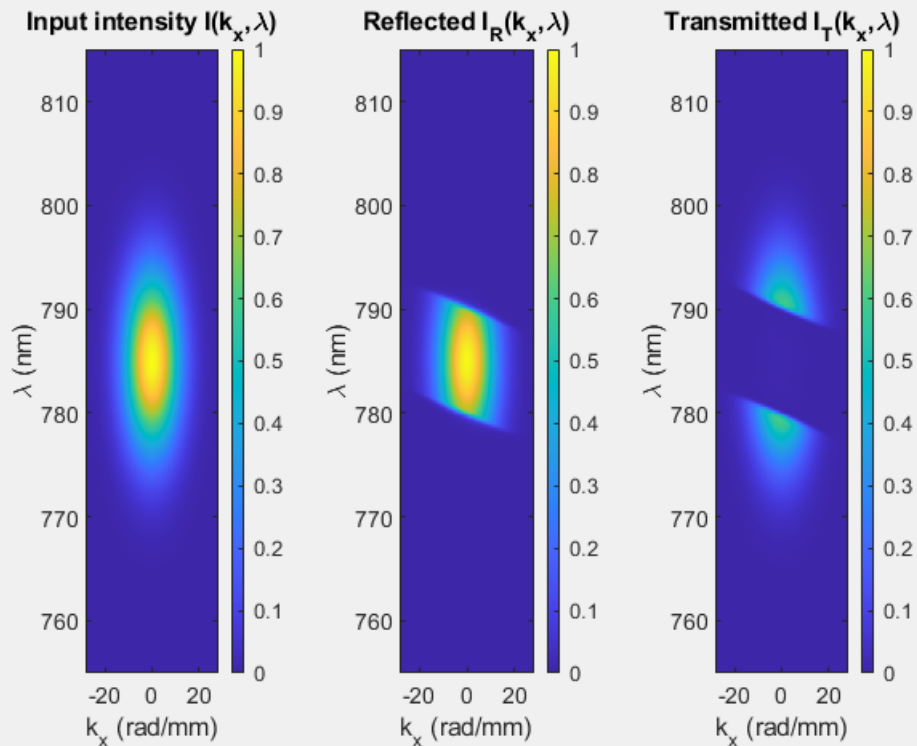


FRED model at 0° AOI

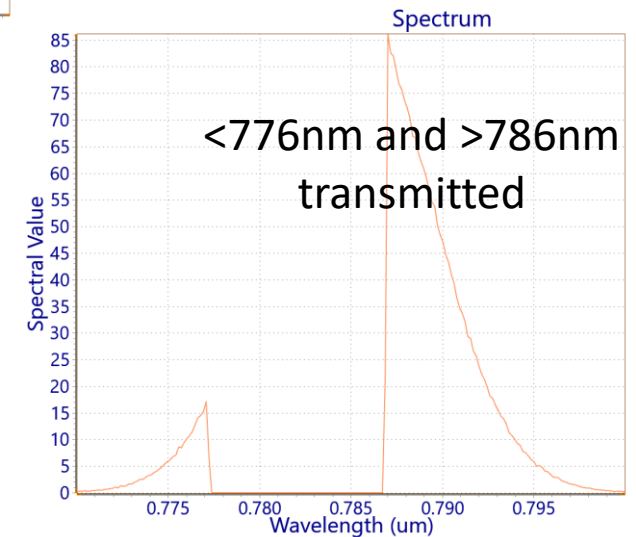
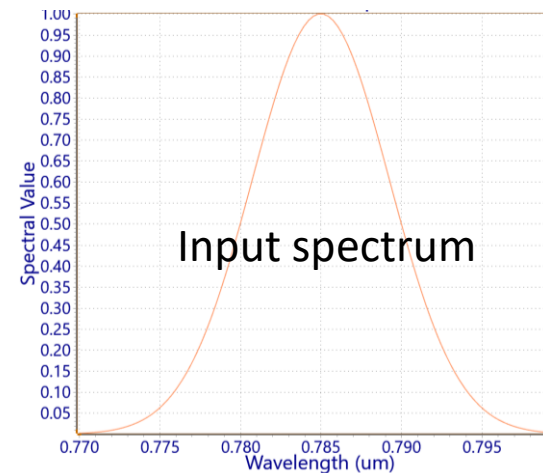
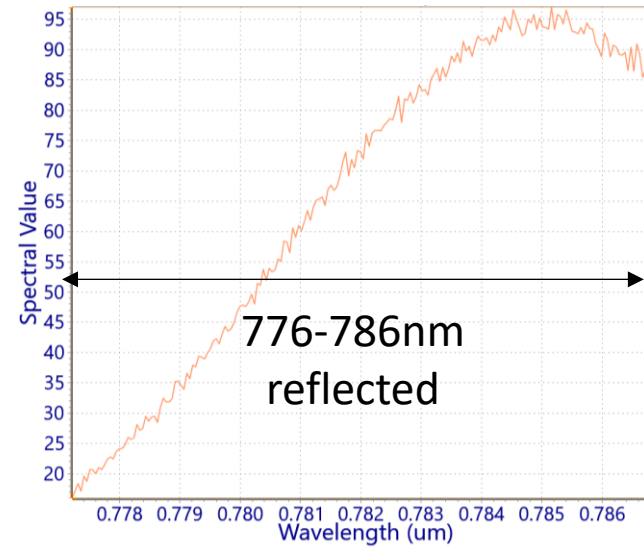


MATLAB model

Spectral domain (k_x, λ)

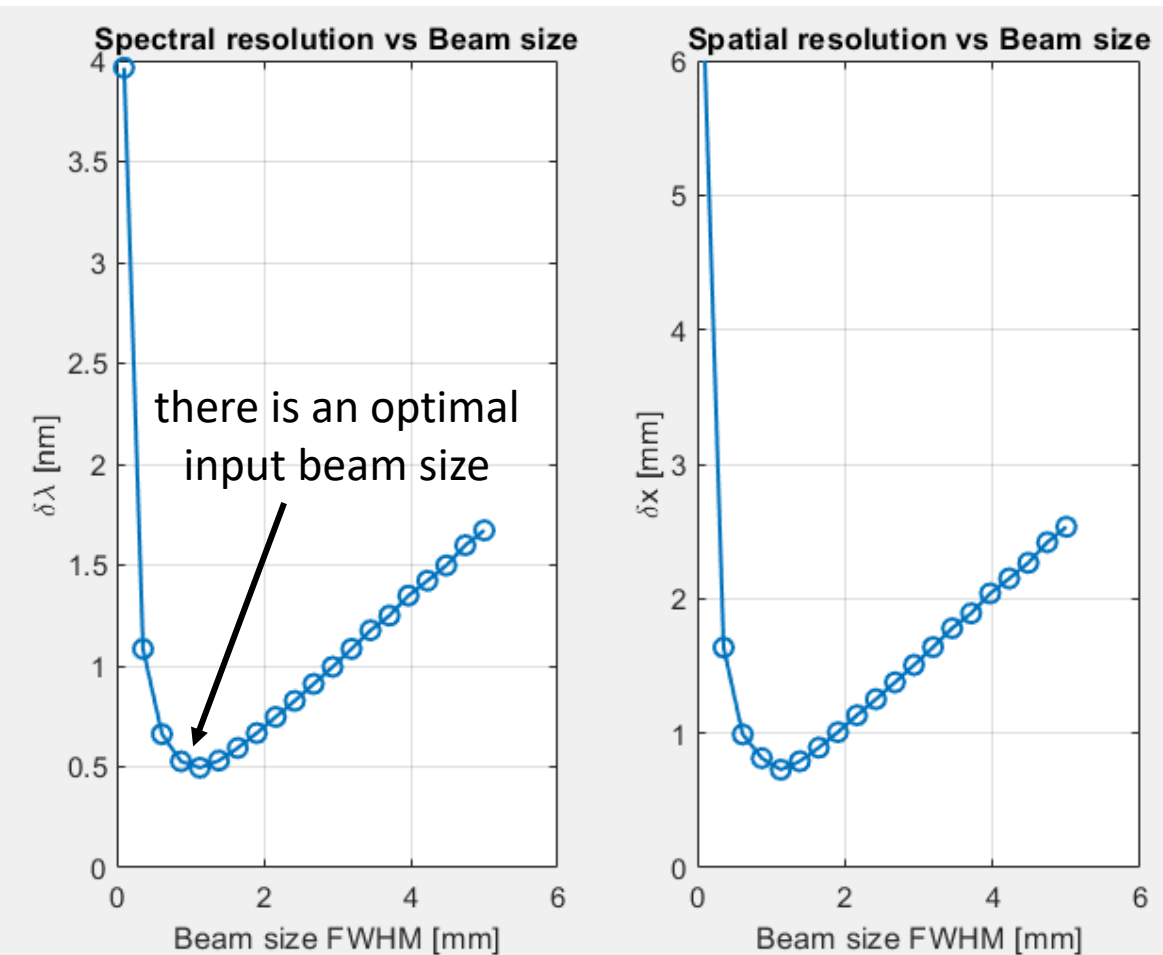


FRED model at -0.3° AOI

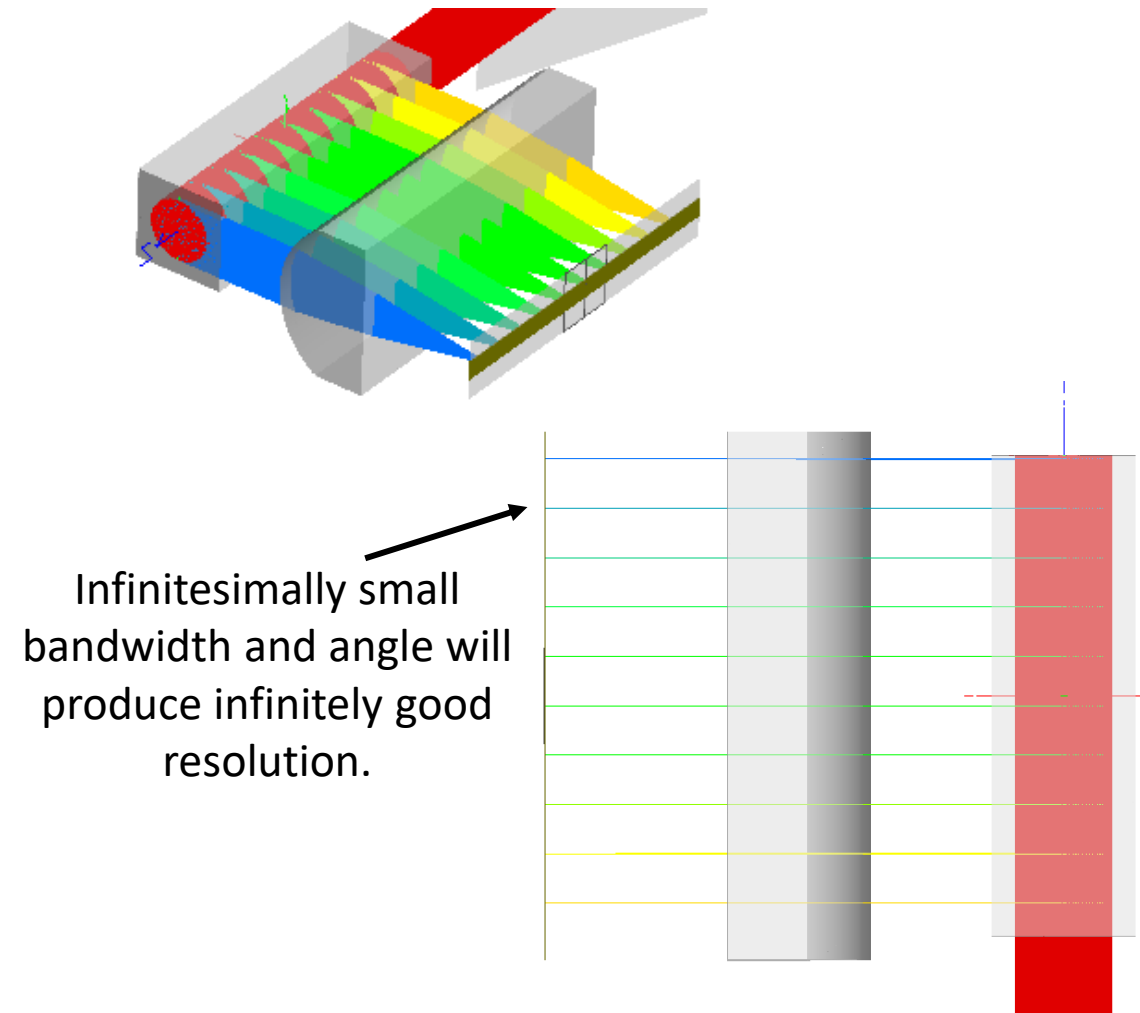


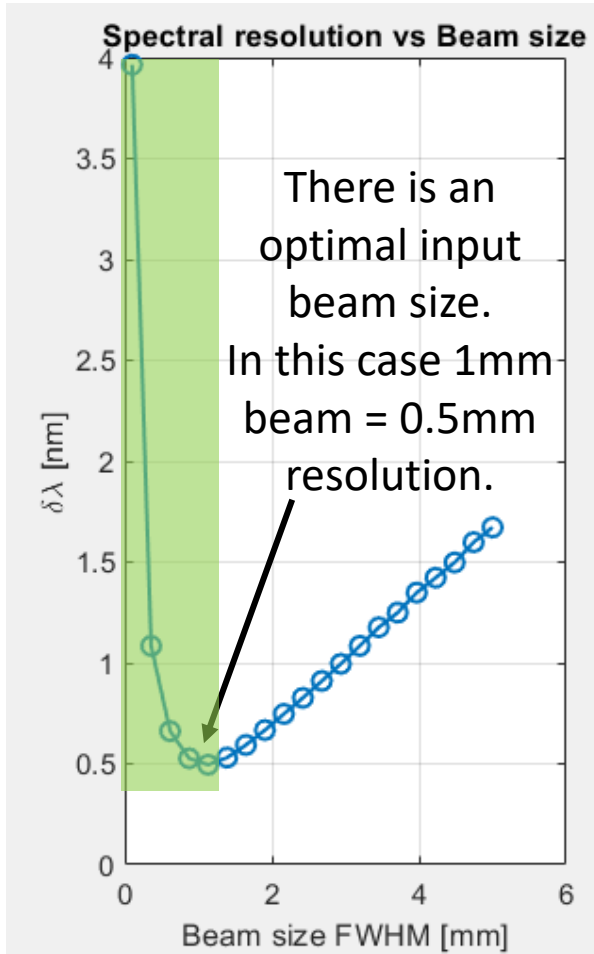
Behaviour of an input beam with varying diameter.

MATLAB model



FRED model





Input beam size in this simulation is directly related to beam divergence.

In a realistic system the input beam won't be a Gaussian beam but a complex spatial and angular distribution.

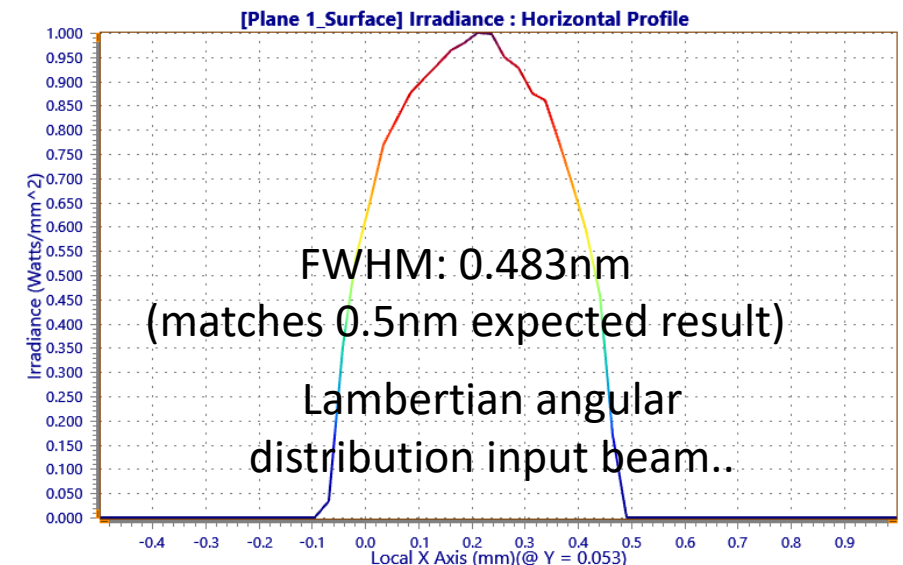
If the optical simulation model around the rCBG is defined according to the physical limitations (etendue / BPP), the script returns realistic results in the green section.

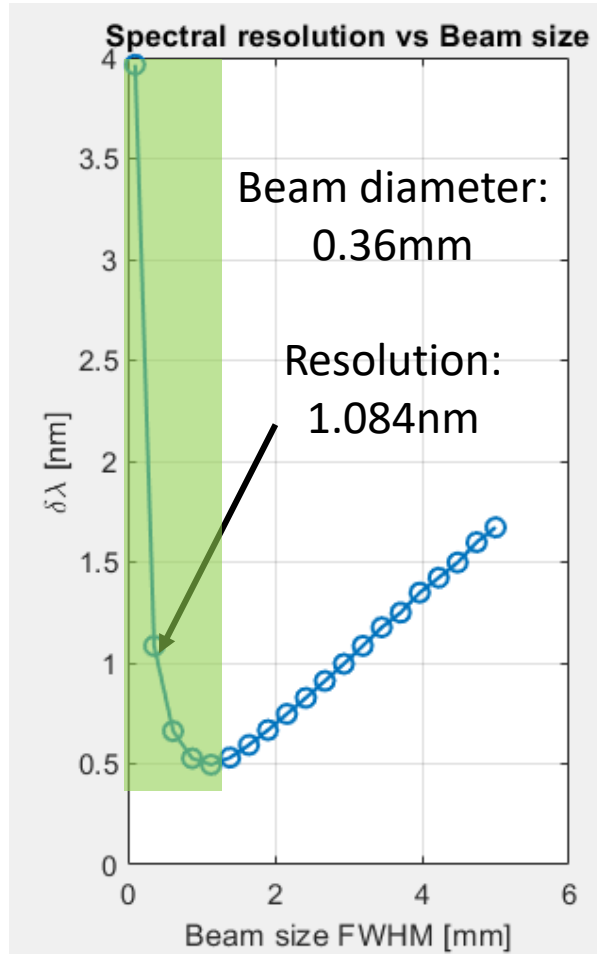
FRED simulation run with 0.028° input beam angle:

Calculator Tool: Laser Beam Quality

Center wavelength:	785 nm	
M^2 factor:	1	calc
Beam parameter product:	0.25 mm mrad	calc
Beam waist radius:	0.5 mm	calc
Divergence half-angle:	500 μrad	calc

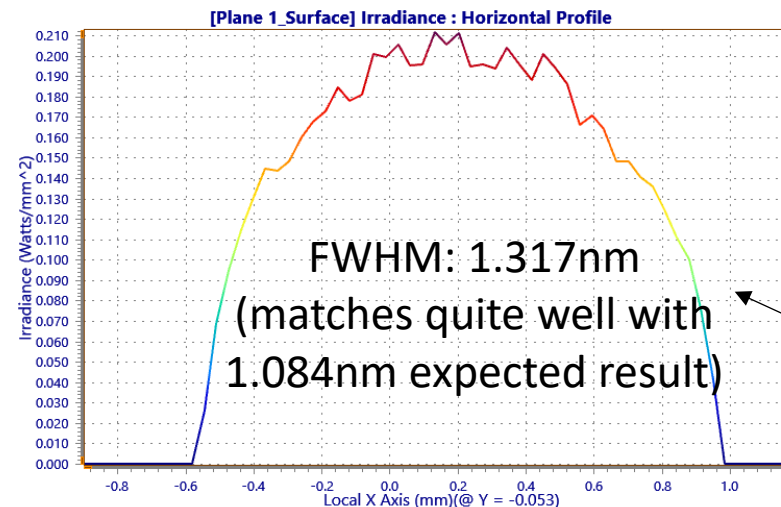
The ideal gaussian beam at 785nm and 1mm beam diameter would have a half-angle divergence of $0.5\text{mrad} = 0.028^\circ$.





Another example

FRED simulation run with
0.08° input beam angle:



Calculator Tool: Laser Beam Quality

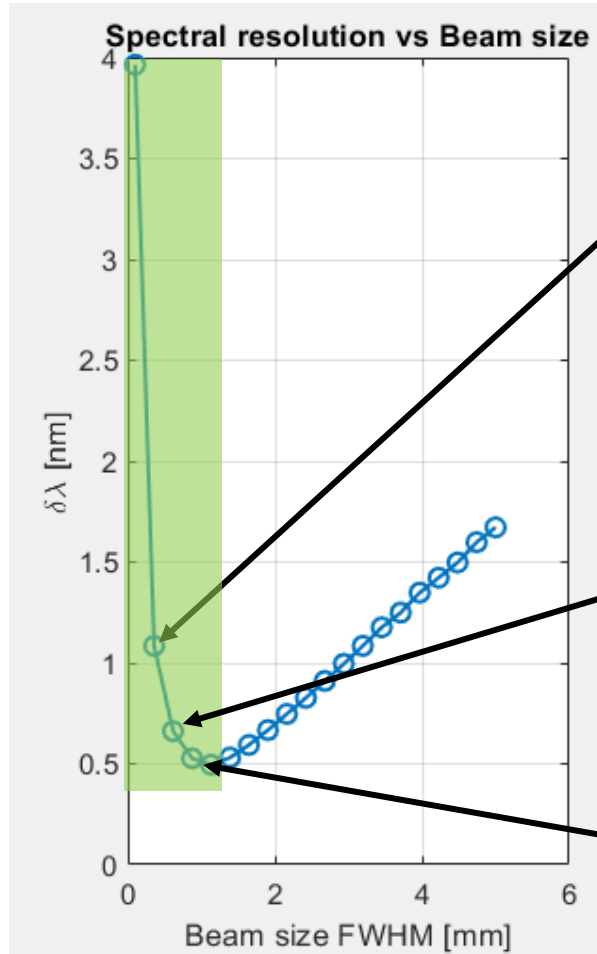
Center wavelength:	785 nm	
M^2 factor:	1	calc (fr
Beam parameter product:	0.25 mm mrad	calc (fr
Beam waist radius:	0.1789mm	calc (u:
Divergence half-angle:	1.4 mrad	calc (u:

The ideal gaussian beam at 785nm and 0.3578mm beam diameter would have a half-angle divergence of 1.4mrad = 0.08°.

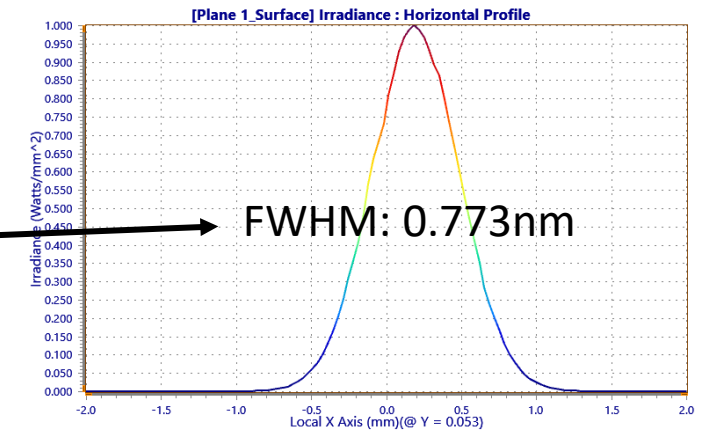
Lambertian angular
distribution input beam..

resolution limit: Gaussian-Gaussian comparison

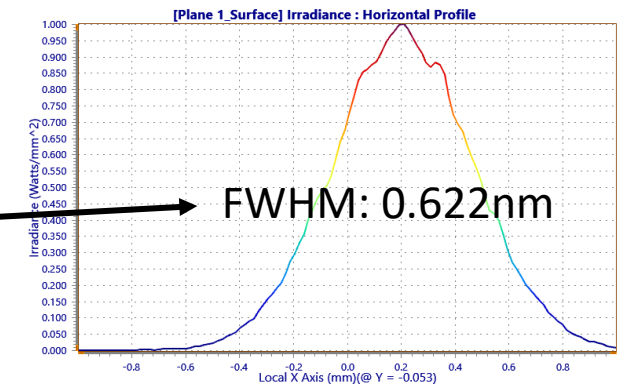
Daniel Csati, 28.04.2026



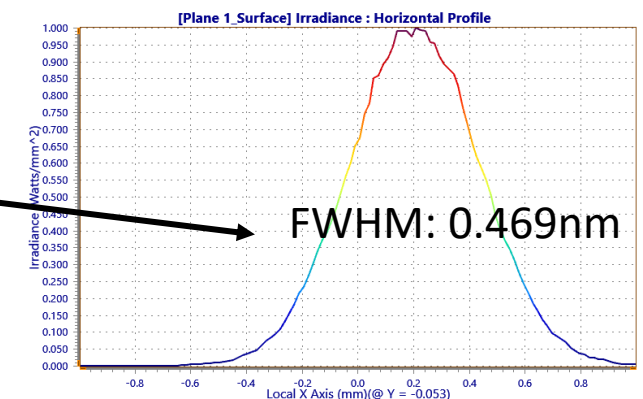
Beam diameter: 0.36mm FWHM
($d=0.61\text{mm}$ @ $1/e^2$, 0.82mrad HAD)
Resolution: 1.09nm



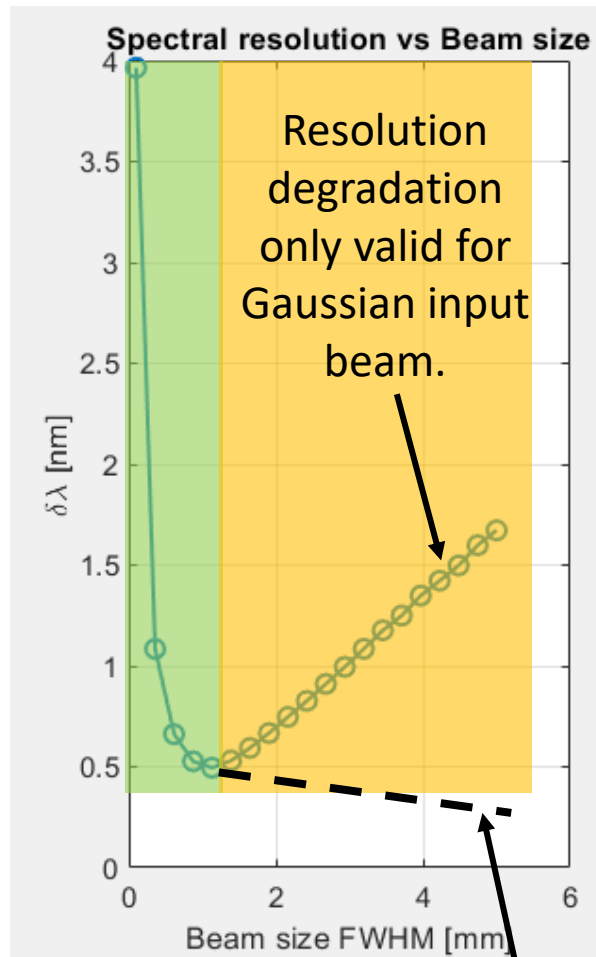
Beam diameter: 0.62mm FWHM
($d=1.05\text{mm}$ @ $1/e^2$, 0.50mrad HAD)
Resolution: 0.66nm



Beam diameter: 1.13mm FWHM
($d=1.92\text{mm}$ @ $1/e^2$, 0.26mrad HAD)
Resolution: 0.50nm



The smaller the beam diameter, the more mismatch in the resolution.



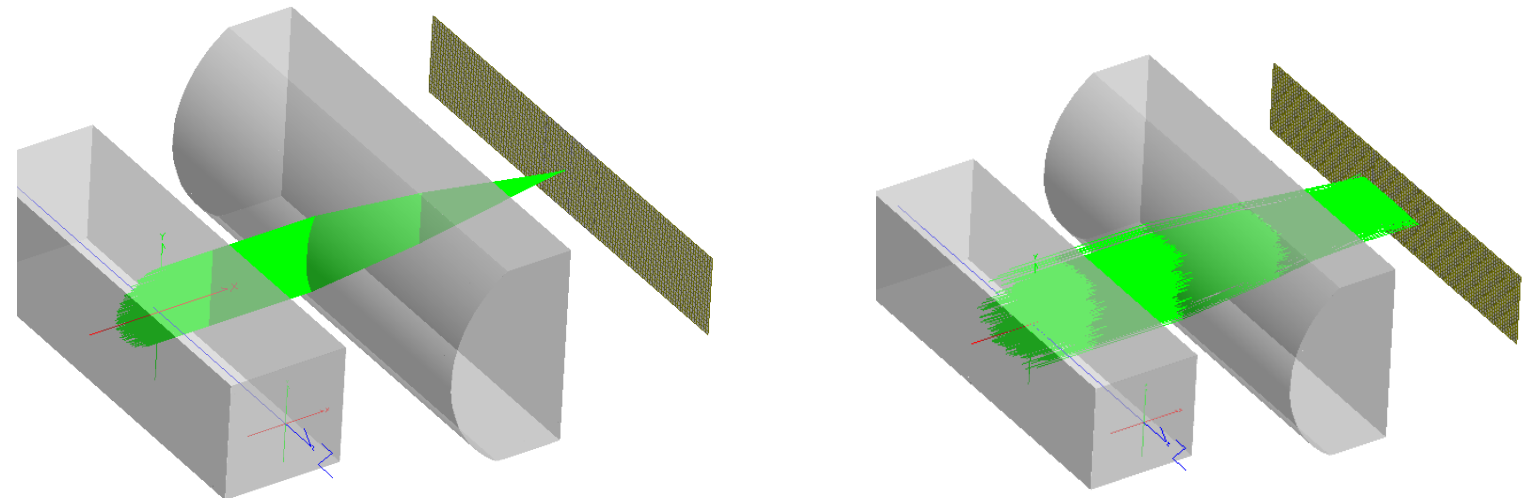
Expected curve from Etendue / BPP.
(Increasing beam diameter = reducing angular spread.)

Implementation:

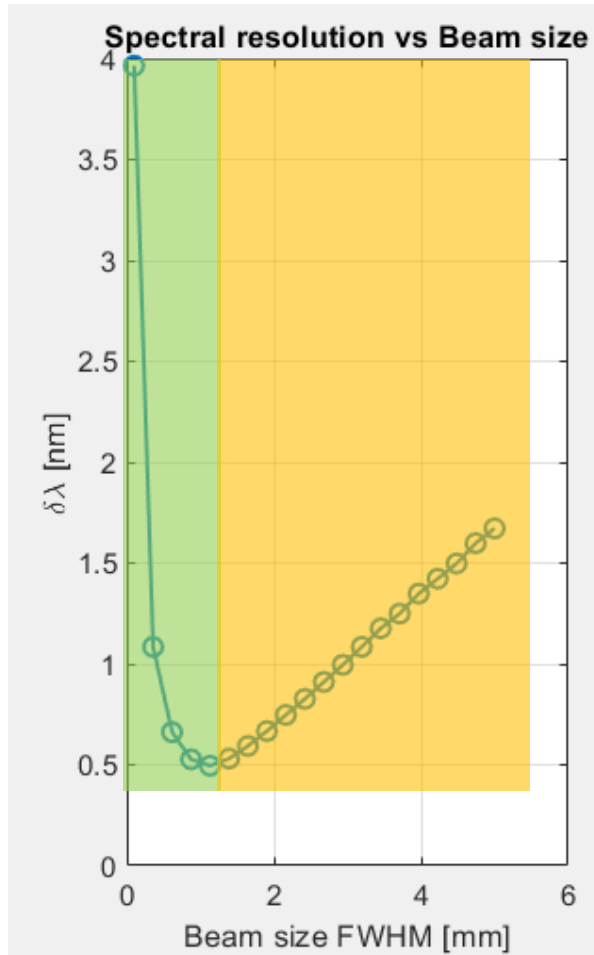
- Measuring the input beam diameter in the running script is unreliable – user input is needed for a chosen beam diameter.
- User input: resolution spread FWHM
- Ray transformation: a +/- Gaussian spread value to the rays wavelength

Important:

- This will not represent real behaviour accurately with non-Gaussian beams.
- In a generally case a complex spatial and angular distribution is expected on the rCBG input face.

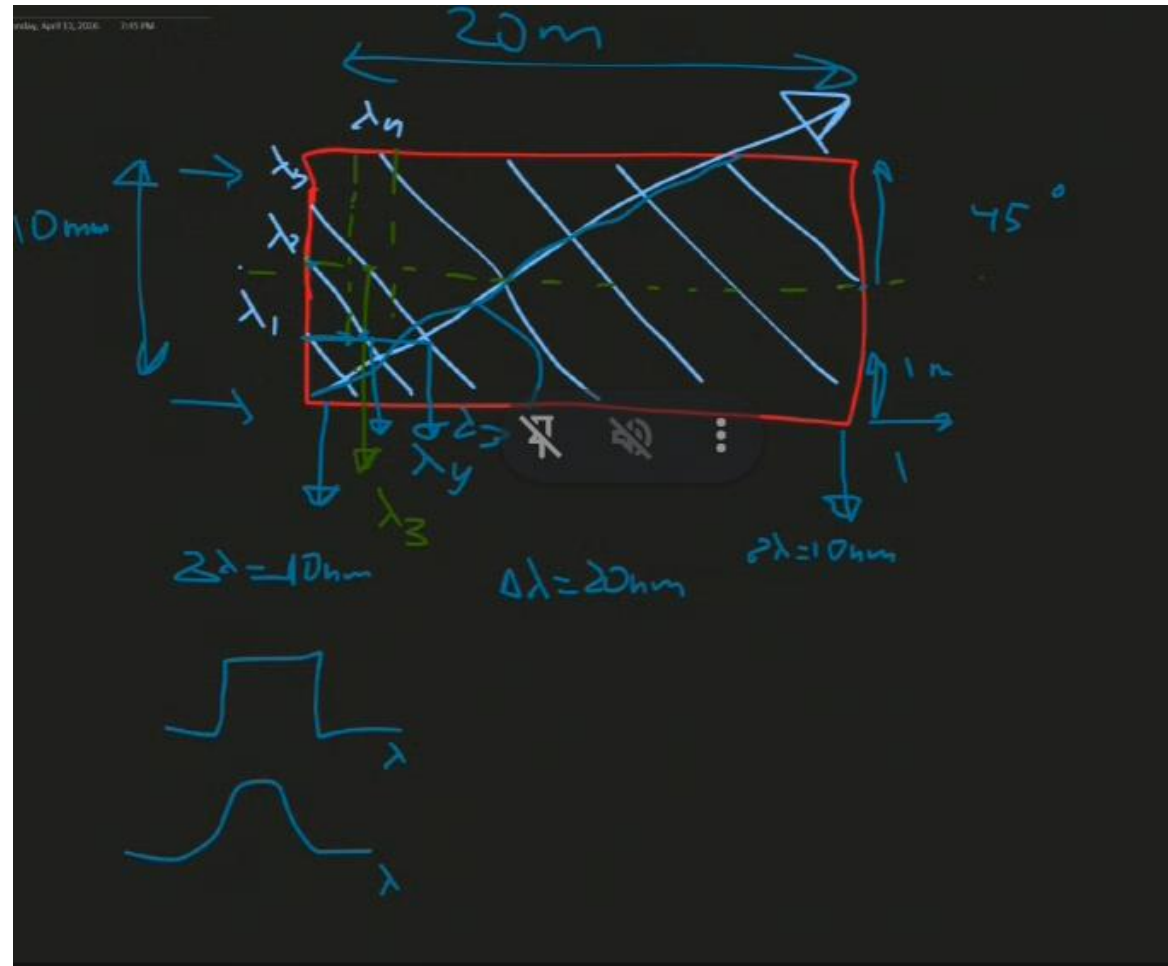


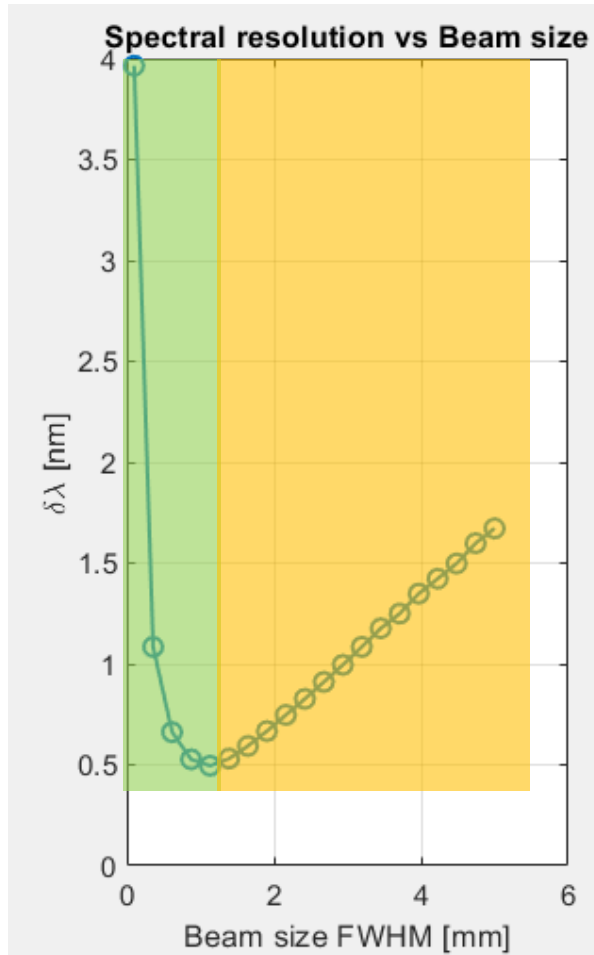
Single wavelength, 0° AOI. Without and with 1nm spectral blur.



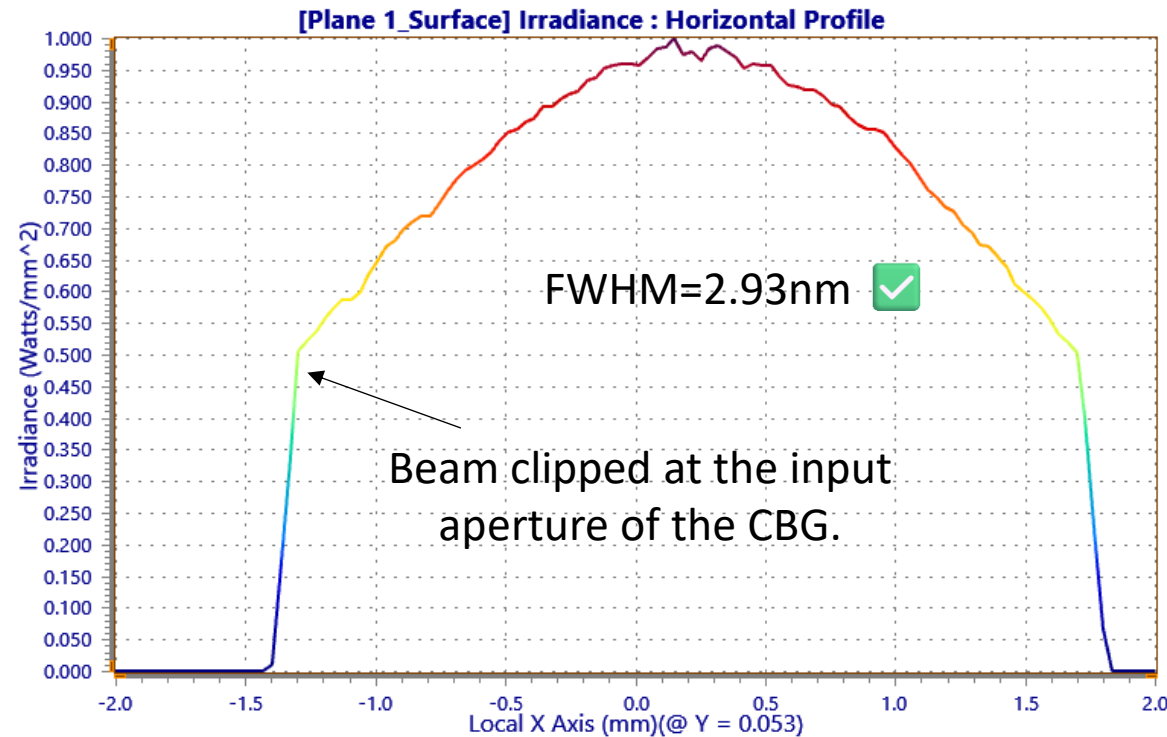
New Implementation:

- 1. Central wavelength and bandwidth is determined by the ray angle. (no change)
- 2. Ray wavelength is offset depending on the input face position offset. Offset towards coupling out side = negative wavelength offset.



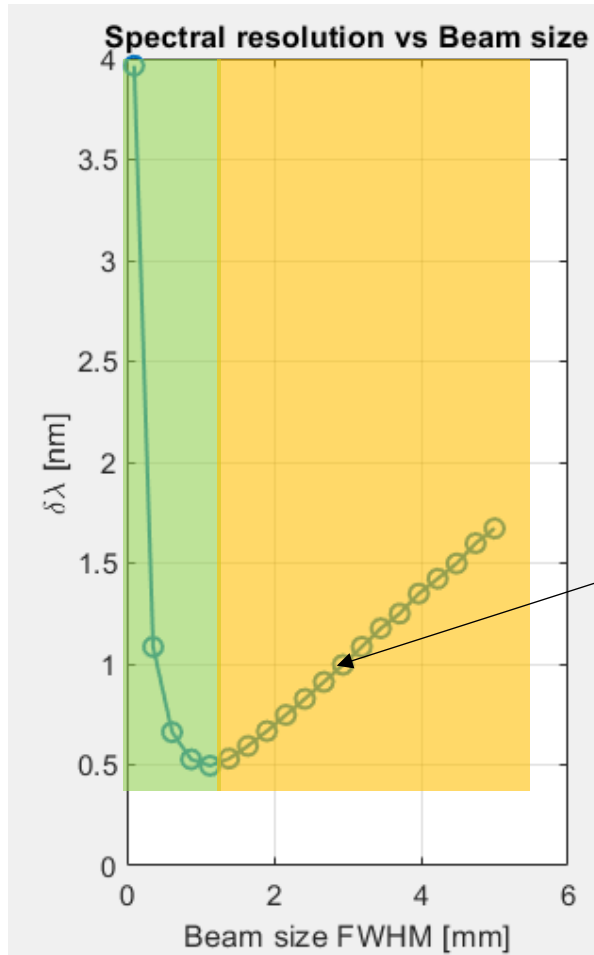


Test: 1nm resolution at 2.93mm FWHM ($d=4.99\text{mm } 1/e^2$).

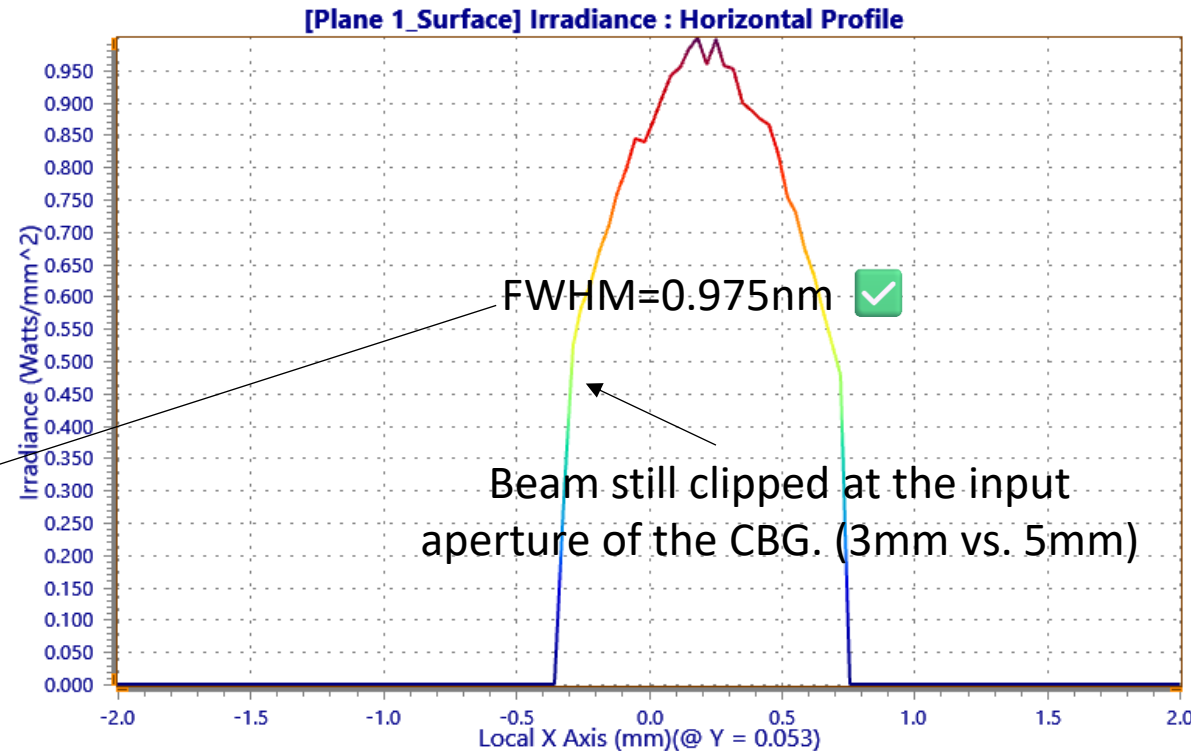


1mm ray height =
1nm offset
(ratio can be changed
in the code)

Input beam gaussian spatial distribution but $\neq 0$ angular distribution. 785nm

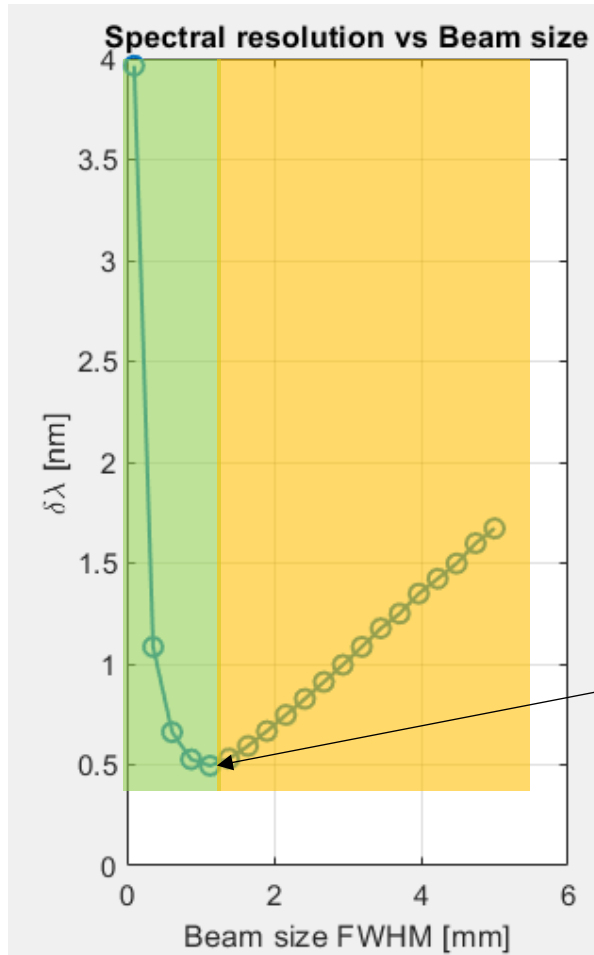


Test: 1nm resolution at 2.93mm FWHM ($d=4.99\text{mm } 1/e^2$).

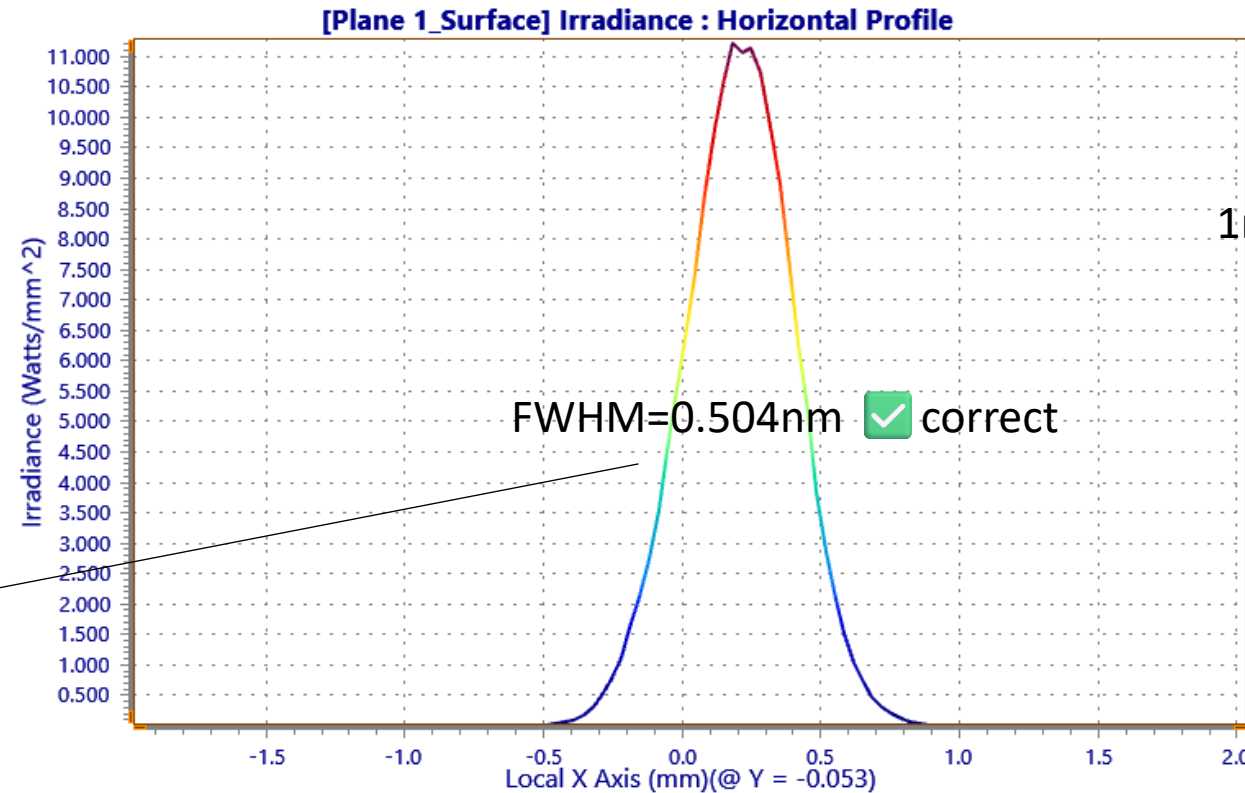


1mm = 0.34nm offset

Input beam gaussian spatial distribution but $\neq 0$ angular distribution. 785nm.



Test: 0.5nm resolution at 1.13mm FWHM ($d=1.92\text{mm}$ @ $1/e^2$, 0.26mrad HAD)



1mm = 1nm offset

Input beam has gaussian spatial distribution and angular distribution.
785nm. (But it's still incoherent.)

- Ray entering on the exit face or any other trace – no tracing in the script
- rCBG block can be placed anywhere in the coordinate system and in any tilt angle
- Rays with one total internal reflection are handled well
- Ray completely out of wavelength range still pass through
- Any source type works (point, gaussian, Lambertian, LED rayfile..)
- Ray exit face can be set + or – direction without issues (if needed) or alternatively the rCBG can be rotated on axis by 180°
- rCBG block size and base refractive index can be changed without issues
- Components around the rCBG can be deleted or changed if not in use

Project target

- Create rCBG model in the first chosen software ✓
- Test model with basic setups ✓

Attachments:

FRED model, script file, work instructions document, MATLAB scripts for reference, results summary slides

Future work

- implementation of the rCBG in Zemax Opticstudio
- analysis of real-world examples (complex optical assemblies) where the system is well known and both rCBG and spectrometer-measurements (reference) of the emission spectra are available